Research Proposal

MSc Thesis in Data Science and Business Analytics

Multi-Objective Optimization for Cloud Architecture Selection on Google Cloud Platform

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1 Introduction

Organizations increasingly rely on large-scale data processing infrastructures to support analytics and machine learning workloads. Cloud computing platforms offer flexible and scalable infrastructure for such workloads, but selecting an appropriate architecture remains a complex decision problem involving multiple trade-offs.

At ING's Wholesale Banking Analytics department, many core data assets are currently processed using an on-premises Data Analytics Platform based on technologies such as Apache Airflow, Apache Spark, and Apache Iceberg. As part of ING's technology strategy, these workloads are being migrated from the internal Private Cloud (IPC) to the Google Cloud Platform (GCP) in order to improve scalability and support new advanced analytics and generative AI use cases.

However, this migration introduces a new architectural decision problem. Tools and configurations that were optimal in the private cloud may no longer be optimal in GCP, and many new architecture alternatives must be considered, consisting of different combinations of orchestration services and processing frameworks (e.g., Cloud Composer, Vertex Pipelines, or Cloud Run Jobs used to orchestrate workflows, combined with processing frameworks such as Polars or BigQuery). This raises the key question of how to systematically identify which architecture is most suitable for a given set of workloads.

A manual approach to this problem would be to define a set of representative workloads and evaluation metrics, run each workload across the different architecture alternatives, and then manually analyze the results to identify a suitable solution. Clearly, such a procedure becomes difficult to scale and lacks a formal decision-making framework, especially when multiple conflicting criteria such as cost, processing time, resource usage, and operational complexity must be considered simultaneously.

Prior research has shown that optimization techniques can support complex design decisions in cloud environments. In particular, multi-objective optimization approaches are widely used to support decision-making in cloud systems. Much of the existing literature focuses on problems such as cloud service selection, deployment allocation, or performance benchmarking (Li et al. 2011; Jahani and Khanli 2016; Luo et al. 2012). Another line of research applies optimization methods to configuration tuning, where parameters of a given system (such as Apache Spark settings) are optimized to improve performance (Cheng et al. 2021; Öztürk 2024). These studies demonstrate that optimization techniques can effectively improve system performance, but they typically assume that the underlying architecture is fixed and focus on optimizing parameters or allocations within an existing architecture.

A smaller body of work formulates cloud infrastructure design problems using formal optimization models. For example, Pérez et al. (2026) model cloud system design as a multi-objective optimization problem and use heuristic search techniques to identify efficient configurations. Similarly, Ciavotta et al. (2017) develop a mixed-integer linear programming (MILP) model to optimize application deployment across multiple cloud providers, showing that mathematical programming can effectively support cloud infrastructure decision-making. However, these approaches still generally focus on resource allocation or configuration choices within a predefined architecture, such as determining optimal deployment strategies or capacity allocations.

In contrast, the higher-level problem of selecting between alternative architectural configurations themselves (such as different orchestration services or data processing frameworks) remains relatively underexplored. This thesis addresses this gap by modeling cloud architecture selection as a multi-objective mixed-integer optimization problem. Architecture components such as orchestration services and processing frameworks will be represented as discrete decision variables, while empirical performance and cost measurements obtained from representative workloads will be incorporated as model parameters. The model will determine Pareto optimal configurations that balance cost, execution time, and operational complexity, subject to resource and service level constraints.

Moreover, if uncertainty in workload characteristics or cost estimates proves substantial, sensitivity analysis will be explored and, where appropriate, robust optimization techniques will be considered to evaluate the stability of selected configurations.

In doing so, this research aims to provide a mathematically grounded and reproducible decision-support framework that identifies optimal or Pareto-optimal configurations based on explicit criteria and constraints, rather than manual inspection alone. Moreover, the structure of the optimization model can be reused and updated as new workloads, tools, or cloud services are introduced, making it a scalable and future-proof approach to cloud architecture selection.

2 Research questions

How can a multi objective mixed integer optimization model support cloud architecture selection in ING's Google Cloud environment by balancing cost, execution time, and operational complexity under performance and resource constraints?

3 Methodology and Techniques

The research will combine empirical experimentation in a cloud environment with mathematical optimization modeling.

First, a set of representative data-processing workloads will be implemented and executed on different architecture configurations within ING's Google Cloud environment (VISTA). These experiments will evaluate combinations of orchestration services and processing frameworks, such as Cloud Composer, Vertex Pipelines, Cloud Run Jobs, BigQuery, and Polars. During these experiments, key performance metrics will be collected, including compute usage (e.g., vCPU and memory consumption), execution time, storage usage, estimated operational cost. These measurements will form the empirical basis for the modeling stage.

Next, the architecture selection problem will be formulated as a multi-objective mixed-integer linear programming (MILP) model. Architectural components will be represented as discrete decision variables, while the observed performance and cost metrics will be incorporated as model parameters. The model will optimize multiple objectives simultaneously, including minimizing infrastructure cost, minimizing processing time, minimizing operational complexity. The resulting optimization problem will be solved to obtain Pareto-optimal architecture configurations, providing a structured decision-support framework for selecting cloud architectures.

Finally, sensitivity analysis will assess how changes in workload characteristics and cost parameters affect the optimal architecture, with robust optimization considered if uncertainty proves significant.

4 Data and/or expected results

The empirical data used in this thesis will come from experimental workloads executed within ING's Google Cloud environment (VISTA), which provides infrastructure for testing analytics pipelines. For each architecture configuration, performance metrics such as compute usage, processing time, storage consumption, and estimated operational costs will be collected. These measurements will serve as input parameters for the optimization model.

The expected outcome of the research is a reproducible optimization-based framework that identifies optimal cloud architecture configurations under multiple objectives and operational constraints. From a practical perspective, the research will provide an evidence-based proposal for a target architecture for ING's analytics workloads in Google Cloud. From an academic perspective, the thesis will demonstrate how mathematical optimization techniques can be applied to support complex infrastructure design decisions in cloud computing environments.

5 Literature

Cheng, Guoli, Shi Ying and Bingming Wang (2021). 'Tuning Configuration of Apache Spark on Public Clouds by Combining Multi-objective Optimization and Performance Prediction Model'. In: *Journal of Systems and Software* 180, p. 111028. DOI: 10.1016/j.jss.2021.111028.

Ciavotta, Michele, Danilo Ardagna and Giovanni Paolo Gibilisco (2017). 'A Mixed Integer Linear Programming Optimization Approach for Multi-cloud Capacity Allocation'. In: *Journal of Systems and Software* 123, pp. 64–78. DOI: 10.1016/j.jss.2016.10.001.

Jahani, Arezoo and Leyli Mohammad Khanli (2016). 'Cloud Service Ranking as a Multi Objective Optimization Problem'. In: *The Journal of Supercomputing* 72.5, pp. 1897–1926. DOI: 10.1007/s11227-016-1690-2.

Li, Jim Zw, Murray Woodside, John Chinneck and Marin Litoiu (2011). 'CloudOpt: Multi-goal Optimization of Application Deployments across a Cloud'. In: *2011 7th International Conference on Network and Service Management*, pp. 1–9.

Luo, Chunjie, Jianfeng Zhan, Zhen Jia, Lei Wang, Gang Lu, Lixin Zhang, Cheng-Zhong Xu and Ninghui Sun (2012). 'CloudRank-D: Benchmarking and Ranking Cloud Computing Systems for Data Processing Applications'. In: *Frontiers of Computer Science* 6.4, pp. 347–362. DOI: 10.1007/s11704-012-2118-7.

Öztürk, M. Maruf (2024). 'Tuning Parameters of Apache Spark with Gauss–Pareto-based Multi-objective Optimization'. In: *Knowledge and Information Systems*. DOI: 10.1007/s10115-023-02032-z.

Pérez, M., Pablo C. Cañizares and Alberto Núñez (2026). 'Multi-objective Optimization of Cloud Systems'. In: *Science of Computer Programming* 252, p. 103447. DOI: 10.1016/j.scico.2026.103447.

6 Timetable

1st Month (April)

Activity	Date
Familiarization with ING analytics infrastructure and Google Cloud (VISTA)	1.
Define representative workloads and evaluation metrics	
Prepare and preprocess test data	
Initial literature review	2.

2nd Month (May)

Activity	Date
Conduct experiments and collect empirical performance metrics	3.
Analyze experimental results	
Formulate the optimization model	
Continue literature review	4.

3rd Month (June)

Activity	Date
Finalize optimization model	5.
Compute Pareto-optimal architecture configurations	
Consider sensitivity analysis and explore robust optimization	
Finalize thesis writing	6.

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Supervisor
J.A.S. Gromicho

Signature ____________________

Second marker (suggestion by supervisor)
Y. Zheng

Thesis coordinator
C.G.H. Diks

Signature
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